



Choice of the reduction and of the representation in centrifugal distortion analysis: A case study of dimethylsulfoxide

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ABSTRACT

The millimeter-wave and submillimeter-wave spectra of dimethylsulfoxide have been measured up to 660 GHz, corresponding to maximum values of $J = 70$ and $K_c = 68$. A total of 3331 transitions (corresponding to 2133 distinct frequencies) were fit to a standard Watson Hamiltonian using the S - and A -reductions and the representations I' and III' . Although dimethylsulfoxide is an asymmetric oblate top, the combination (A, III') gives the worst results. From the point of view of the convergence of the Hamiltonian, the best results are obtained with the combination (S, III') but the least-squares system of normal equations is better conditioned with representation I' . These results are compared to those obtained for a few typical molecules.

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1. Introduction

The rovibrational Hamiltonian as reduced by Watson has been highly successful to analyze molecular spectra [1]. For the analysis of a non-degenerate vibrational state of a symmetric or asymmetric top, two reductions are in use. The A (asymmetric)-reduction is generally used for asymmetric tops and the S (symmetric)-reduction is preferred for molecules close to the symmetric top. Furthermore, there are six different ways (representations) to identify the (x, y, z) reference system with the (a, b, c) principal axis system. In practice two different representations are used: I' where $x = b, y = c, z = a$ and which is best for prolate molecules (Ray's asymmetry parameter $\kappa = (2B - A - C)/(A - C) < 0$) and III' where $x = a, y = b, z = c$ (representation III' is also used where $x = a, y = c, z = b$ but it is equivalent to representation III') and which is thought to be better for oblate molecules ($\kappa > 0$). However, the choice of the right reduction and of the right representation is not always obvious. Different reductions and representations have been tried on many molecules but there are few critical studies. It is often assumed that the S -reduction is always as good as, or better than, the A -reduction whereas the III'^{II} representation is believed to be more appropriate for oblate molecules. However, some authors prefer to use the I' representation for oblate molecules, one typical example being the ground state rotational spectrum of nitric acid, HNO_3 [2,3].

One exception to this relative lack of critical studies is the analysis of the ground state rotational spectrum of the oblate molecule carbonyl fluoride, OCF_2 (asymmetry parameter, $\kappa = +0.980$). Carpenter [4] compared the different representations together with the A -reduction and found that the convergence was poorer for representation III' . Later, Typke [5] compared the different reductions and found better results with the S -reduction. In the same work, he also analyzed the ground state rotational spectrum of dimethyl sulfoxide (DMSO), $(\text{CH}_3)_2\text{SO}$. Finally, in a more systematic comparative studies, Cohen and Lewis-Bevan [6] could demonstrate that, for the ground state, any of the combinations (S, III') , (A, I') or (S, I') gives excellent fits of the combined experimental data (microwave measurements and infrared data). On the other hand, the combination (A, III') is clearly inappropriate since the fit of the experimental data is about 100 times worse than for the other choices. DMSO is particularly interesting to perform such a comparative study because it is an oblate molecule not too far from the symmetric-top limit (asymmetry parameter, $\kappa = +0.886$) and its rotational spectrum is intense with a great variety of lines. Furthermore, it was the first non-planar molecule subjected to a centrifugal distortion analysis.

The microwave spectrum of DMSO was first investigated by Dreizler and Dendl [7] who determined the rotational constants and the electric dipole moment ($\mu_b = 3.94 \text{ D}$ and $(\mu_b = 0.40 \text{ D})$ of the vibrational ground state. A little bit later, the same authors [8] tried to determine the six quartic centrifugal distortion constants using the first-order Hamiltonian of Kivelson and Wilson [9]. Although they could reproduce well the experimental frequencies, the derived centrifugal distortion constants were extremely inaccurate because the system of normal equations is highly

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ill-conditioned. This difficulty was solved by Watson [10,11] who showed that there is a linear relation between the average values of the angular-momentum operators that appear in the equation of Kivelson and Wilson. This is the first time that a severe correlation problem was encountered in vibration–rotation spectroscopy. This late appearance is simply due to the fact that the centrifugal distortion analyses were limited to planar molecules where the number of independent quartic centrifugal distortion constants is only four. The analysis of a non-planar molecule required a powerful computer which was rarely available before the 1960s.

The five quartic centrifugal distortion constants of the ground state of DMSO were subsequently determined by Typke [12] and improved by Fliege et al. [13]. The latter authors also determined the potential barrier to internal rotation. The substitution (r_s) structure was calculated by Feder et al. [14] and improved by Typke [15]. The infrared spectrum was measured by Geiseler and Hanschmann [16] and a reliable harmonic force field was determined by Typke and Dakkouri [17] by scaling an *ab initio* force field with the help of experimental data. In the same work, an average (r_z) structure was calculated using microwave rotational constants and electron diffraction data together with the new force field.

The present paper is organized as follows. Section 2 summarizes the theoretical background. In Section 3, the calculation of the *ab initio* harmonic force field is explained. Section 4 describes the measurement of the spectrum of DMSO while Section 5 focuses on its analysis. Finally, in Section 6, the fits obtained on DMSO are compared to those of a few other typical molecules.

2. Theory

According to Watson [1] the rotational Hamiltonian for an asymmetric rotor is expanded as a function of the angular-momentum operators J_x^2, J_z^2 and $J_x \pm J_y$. It may be written

$$\mathbf{H} = \mathbf{H}_2 + \mathbf{H}_4 + \mathbf{H}_6 + \dots \quad (1)$$

with

$$\mathbf{H}_2 = \frac{1}{2}(B_x + B_y)\mathbf{J}^2 + \frac{1}{2}(2B_z - B_x - B_y)J_z^2 + \frac{1}{4}(B_x - B_y)(J_+^2 + J_-^2) \quad (2a)$$

$$\mathbf{H}_4 = T_{400}\mathbf{J}^4 + T_{220}J_z^2\mathbf{J}^2 + T_{040}J_z^4 + T_{202}(J_+^2 + J_-^2)\mathbf{J}^2 + \frac{1}{2}T_{022}\{J_z^2, (J_+^2 + J_-^2)\} + T_{004}(J_+^4 + J_-^4) \quad (2b)$$

$$\mathbf{H}_6 = \Phi_{600}\mathbf{J}^6 + \Phi_{420}J_z^2\mathbf{J}^4 + \Phi_{240}J_z^4\mathbf{J}^2 + \Phi_{060}J_z^6 + \Phi_{402}(J_+^2 + J_-^2)\mathbf{J}^4 + \frac{1}{2}\Phi_{222}\{J_z^2, (J_+^2 + J_-^2)\}\mathbf{J}^2 + \frac{1}{2}\Phi_{042}\{J_z^4, (J_+^2 + J_-^2)\} + \Phi_{204}\mathbf{J}^2(J_+^4 + J_-^4) + \frac{1}{2}\Phi_{024}\{J_z^2, (J_+^4 + J_-^4)\} + \Phi_{006}(J_+^6 + J_-^6) \quad (2c)$$

where the $\{A, B\}$ is the anticommutator $AB + BA$.

Watson [1,18,19] has shown that of the six quartic distortion constants T , only five combinations are determinable from the spectra and only seven combinations of the 10 sextic constants can be obtained from the spectra. To obtain this result, Watson subjected the original Hamiltonian to a unitary transformation (which does not alter the eigenvalues)

$$\tilde{\mathbf{H}} = e^{-i\mathbf{S}}\mathbf{H}e^{i\mathbf{S}} \quad (3)$$

where the first term of the transformation operator is written

$$\mathbf{S} = s_{111}(J_x J_y J_z + J_z J_y J_x) \quad (4)$$

Watson furthermore showed that the parameters s_{111} should not exceed the order of magnitude T/B in order to insure a fast convergence of the Hamiltonian. T and B are the typical values of the quartic distortion constants and the rotational constants for the molecule in question. To fix the free parameter s_{111} , Watson proposed two sets of constraints:

$$\text{A-reduction : } \tilde{T}_{004} = \tilde{\Phi}_{204} = \tilde{\Phi}_{024} = \tilde{\Phi}_{006} = 0 \quad (5a)$$

$$\text{S-reduction : } \tilde{T}_{022} = \tilde{\Phi}_{222} = \tilde{\Phi}_{042} = \tilde{\Phi}_{024} = 0 \quad (5b)$$

The A-reduction implies

$$s_{111} = -\frac{4T_{004}}{B_x - B_y} = -\frac{4R_6}{B_x - B_y} \quad (6a)$$

and the S-reduction

$$s_{111} = \frac{T_{022}}{2B_z - B_x - B_y} = \frac{2R_5}{2B_z - B_x - B_y} \quad (6b)$$

For each reduction scheme considered, the determinable parameters are effective values which are combinations of rotational and centrifugal distortion constants. The parameters defining the reduction transformation of the Hamiltonian have to arise in the correct order of magnitude [20]. Otherwise, the Hamiltonian does not converge properly which leads to the need to include a large number of higher order parameters, resulting in problems with correlations between the parameters and slow convergence of the fit.

The Hamiltonian in A-reduction may be written

$$\begin{aligned} \mathbf{H}^{(A)} = & B_x^{(A)}J_x^2 + B_y^{(A)}J_y^2 + B_z^{(A)}J_z^2 - \Delta_J(J^2)^2 - \Delta_{JK}J^2J_z^2 - \Delta_KJ_z^4 \\ & - \frac{1}{2}\{\delta J^2 + \delta_KJ_z^2, J_+^2 + J_-^2\} + \Phi_J(J^2)^3 + \Phi_{JK}(J^2)^2J_z^2 \\ & + \Phi_{KJ}J_z^4 + \Phi_KJ_z^6 + \frac{1}{2}\{\varphi_J(J^2)^2 + \varphi_{JK}J^2J_z^2 + \varphi_{KJ}J_z^4, J_+^2 + J_-^2\} \\ & + L_{JK}(J^2)^3J_z^2 + L_{JK}(J^2)^2J_z^4 + L_{KKJ}J^2J_z^6 + L_KJ_z^8 + \dots \end{aligned} \quad (7)$$

whereas the expression of the Hamiltonian in S-reduction is

$$\begin{aligned} \mathbf{H}^{(S)} = & B_x^{(S)}J_x^2 + B_y^{(S)}J_y^2 + B_z^{(S)}J_z^2 - D_J(J^2)^2 - D_{JK}J^2J_z^2 - D_KJ_z^4 \\ & + d_1J^2(J_+^2 + J_-^2) + d_2(J_+^4 + J_-^4) + H_J(J^2)^3 + H_{JK}(J^2)^2J_z^2 \\ & + H_{KJ}J_z^4 + H_KJ_z^6 + h_1(J^2)^2(J_+^2 + J_-^2) + h_2J^2(J_+^4 + J_-^4) \\ & + h_3(J_+^6 + J_-^6) + L_{JK}(J^2)^3J_z^2 + L_{JK}(J^2)^2J_z^4 + L_{KKJ}J^2J_z^6 + L_KJ_z^8 \\ & + \dots \end{aligned} \quad (8)$$

The first criterion to be used in order to choose a reduction and a representation is the value of s_{111} which should be as small as possible and smaller than T/B . The second criterion is obviously the values of the residuals of the fit which should be of the same order of magnitude as the experimental accuracy and which should not show any systematic deviations (i.e. they should have a random behavior). As a first approximation, the standard deviation of the fit may be used but a detailed inspection of the residuals remains indispensable. These two criteria are not sufficient. It is also necessary to check that the fitted parameters (i.e. the centrifugal distortion constants) are precisely determined (the standard deviation of the parameter should be smaller than the value of the parameter) and have the right order of magnitude. Finally, it is also extremely useful to check that the system of normal equations is not ill-conditioned (see Appendix A for the definition of the condition number).

3. *Ab initio* force field

To check the accuracy of the experimental quartic centrifugal distortion constants and to estimate the value of s_{111} , an *ab initio* harmonic force field of DMSO was calculated. The Kohn–Sham density functional theory [21] using Becke's three-parameter hybrid exchange functional [22] and the Lee–Yang–Parr correlation functional [23], together denoted as B3LYP was used. It is indeed now well established that density functional theory provides accurate vibrational frequencies at low cost. Good results are usually obtained with the B3LYP method [24,25]. The split-valence basis set 6–311G, as implemented in GAUSSIAN03 [26], including appropriate polarization functions was used for this calculation. The accuracy of the derived force field was confirmed by comparing it to the scaled force field of Typke and Dakkouri [17]. The value of s_{111} for the different reductions and representations is given in Table 1 and is compared to the values of T_{gg}/B_g where $g = a, b, c$. It appears that the best choice is, as anticipated, (S, III'), closely followed by (S, I') and (A, I'). On the other hand, s_{111} is much larger for (A, III'), which is at first sight a little bit surprising because the representation III' is expected to give better results than the I'-one and DMSO should be asymmetric enough to use the A-reduction. Actually, the representation III' is useful because it minimizes the non-diagonal term in $\mathbf{H}_2$ (last term of Eq. (2a)) for an oblate molecule. However, when the higher-order terms ($\mathbf{H}_4, \mathbf{H}_6, \dots$) are taken into account, this advantage disappears. Whereas the numerators of s_{111} , Eqs. (6) are of the same order of magnitude for the different reductions and representations, the corresponding denominators are hugely different, see Table 1. For instance, $B_x - B_y = 2651$ MHz for the representation I' whereas it is only 160 MHz for the representation III'. The small value of $B_x - B_y$ is an advantage for the rigid rotor ($\mathbf{H}_2$ term only) but it leads to a slow convergence of the Hamiltonian when the centrifugal distortion is taken into account.

4. Experimental details

The millimeter and submillimeter-wave spectra were recorded in the spectral range 150–660 GHz. In the upper frequency range 580–660 GHz, the source is a Russian Istok backward wave oscillator (BWO). It is phase locked on a harmonic from an Agilent (2–20 GHz) synthesizer: a part of the signal from the BWO and the one of an active sextupler from Millitech (75–100 GHz) are mixed on a Schottky planar diode placed in a parabolic structure (from

Table 1

Ab initio rotational parameters for dimethylsulfoxide (from the B3LYP/6+311+G(3df,2p) calculation).

Parameter	Value	T_{gg}/B_g^a
A (MHz)	6970.3	
B (MHz)	6810.5	
C (MHz)	4159.3	
T_{aa} (kHz)	−6.456	-9.3×10^{-07}
T_{bb} (kHz)	−7.312	-1.1×10^{-06}
T_{cc} (kHz)	−1.158	-2.8×10^{-07}
T_{ab} (kHz)	−4.270	
T_{ac} (kHz)	−1.641	
T_{bc} (kHz)	−1.684	
Representation	I'	III'
$2B_x - B_x - B_y$ (MHz)	2970.9	−5462.3
$B_x - B_y$ (MHz)	2651.2	159.8
$2R_5$ (kHz)	0.224	0.192
R_6 (kHz)	−0.319	−0.327
$s_{111}(A)$	4.81×10^{-07}	8.18×10^{-06}
$s_{111}(S)$	7.55×10^{-08}	-3.52×10^{-08}

^a $g = a, b, c$.

Table 2

Rotational parameters in S-reduction for the ground state of dimethyl sulfoxide.

Representation		I'	III'
A	MHz	7036.579634(32)	7036.582334(26)
B	MHz	6910.830104(26)	6910.830327(25)
C	MHz	4218.779498(21)	4218.776781(40)
D_J	kHz	3.463079(14)	6.089201(16)
D_{JK}	kHz	−0.925893(71)	−8.937244(53)
D_K	kHz	3.767553(89)	3.989320(37)
d_1	kHz	−1.4548584(69)	0.164061(11)
d_2	kHz	−0.2939137(42)	−0.2717127(48)
H_J	Hz	−0.0027288(53)	0.0088720(43)
H_{JK}	Hz	0.051058(41)	−0.038929(21)
H_{KJ}	Hz	−0.116535(87)	0.051699(27)
H_K	Hz	0.085683(80)	−0.021331(15)
h_1	Hz	0.0012583(22)	0.0016709(38)
h_2	Hz	0.0026560(23)	0.0010231(23)
h_3	Hz	−0.00011300(68)	0.00134332(85)
L_{JK}	mHz	−0.000121(11)	−0.0000781(49)
L_{KKJ}	mHz	0.000554(24)	0.0001470(85)
L_K	mHz	−0.000633(20)	−0.0000706(43)
s^a	kHz	15.8	18.7
$s(\text{MAD})^b$	kHz	11.0	13.9
Lines ^c		1942	1982
η^d		327	663

^a Unweighted standard deviation of the fit.

^b Standard deviation from the median absolute value, Eq. (A7).

^c Number of transitions retained in the fit.

^d Condition number, Eq. (A2).

Virginia Diodes Inc.). In order to minimize frequency determination errors, the Agilent synthesizer was locked on a GPS reference signal.

In the lower frequency range (150–550 GHz), solid state sources are used. The frequency of the same Agilent synthesizer is first multiplied by six and amplified with a Spacek active sextupler. The output power is +15 dBm in the W-band range (75–110 GHz). This power is high enough to use passive Schottky multiplier (X2, X3, X5) from Virginia Diodes Inc.

The detector is an InSb liquid He-cooled bolometer from Queen Mary College. In order to improve the sensitivity of the spectrom-

Table 3

Rotational parameters in A-reduction for the ground state of dimethyl sulfoxide.

		I'	III'
A	MHz	7036.576681(33)	7036.49312(10)
B	MHz	6910.833844(26)	6910.92455(10)
C	MHz	4218.778084(21)	4218.77257(13)
Δ_J	kHz	4.050907(14)	6.62679(10)
Δ_{JK}	kHz	−4.452825(45)	−12.15739(44)
Δ_K	kHz	6.706545(70)	6.67255(37)
δ_J	kHz	1.4548677(66)	−0.164790(83)
δ_K	kHz	1.285585(18)	−46.8755(53)
Φ_J	Hz	0.0025809(39)	0.006694(29)
Φ_{JK}	Hz	0.017314(23)	−1.17573(35)
Φ_{KJ}	Hz	−0.083466(57)	3.8841(10)
Φ_K	Hz	0.080893(66)	−2.71655(68)
ϕ_J	Hz	0.0011453(19)	0.002600(30)
ϕ_{JK}	Hz	0.0082096(93)	−0.16266(46)
ϕ_K	Hz	0.004977(16)	−0.9056(31)
L_{JK}	mHz	−0.000351(11)	0.08486(23)
L_{KKJ}	mHz	0.001273(25)	−0.24008(64)
L_K	mHz	−0.001134(21)	0.15556(41)
s^a	kHz	16.1	50.8
$s(\text{MAD})^b$	kHz	11.2	45.0
Lines ^c		1945	1817
η^d		225	10 778

^a Unweighted standard deviation of the fit.

^b Standard deviation from the median absolute value, Eq. (A7).

^c Number of transitions retained in the fit.

^d Condition number, Eq. (A2).

with the S- and A-reductions and the representations I' and III'. It has the advantage to be a “robust” method, i.e. it mitigates the influence of outliers, see [Appendix A](#) for the calculation of the weights. Most of the fits were performed with a modified version of the program ASFIT of Z. Kisiel (Z.Kisiel, PROSPE – Programs for ROTational SPEctroscopy, <http://info.ifpan.edu.pl/~kisiel/prospe.htm>).

The results of the fits are given in Table 2 for the S-reduction and in Table 3 for the A-reduction and the measured transitions are given in Table S1 of the supplementary material. As anticipated, the A-reduction in representation III' gives the worst results: the standard deviation of the fit is much larger and the number of excluded lines (i.e. outliers) is the largest one. The condition number of the fit is also extremely large, indicating the importance of ill-conditioning. At first sight, the S-reduction in representation III', is the best choice because the number of rejected lines is the smallest one but both its standard deviation and its condition number are larger than for the representation I' with the S- and A-reduction. It might be argued that it is an artifact because the number of rejected lines is larger in these last two fits. Actually, even when

[illegible]

the same number of lines (with the same weights) is fitted, the order of the standard deviations and of the condition numbers does not change. Furthermore, looking at the residuals of the rejected lines using representation I', it appears that they are quite similar to those found when the S-reduction is used together with the representation III' (where all the lines are included in the fit), see Table 4.

It is also interesting to look at the order of magnitude of the centrifugal distortion constants and more particularly at the diagonal octic constants L_{JK} , L_{KKJ} , and L_K . They are the smallest ones with the combination (S, III'). They are larger with (S, I'), even larger with (A, I') and the largest ones with (A, III'). This is in agreement with the value of s_{111} and confirms that the convergence of the Hamiltonian is the fastest one with the combination (S, III'). However, it cannot be concluded that this combination gives the best fit because the condition number is larger and some parameters are more highly-correlated.

6. Discussion

It is interesting to note that the two oblate molecules COF₂ and DMSO have the same behavior from the point of view of the centrifugal distortion analysis. It would be useful to check whether such a behavior is general for oblate molecules. The case of furane, $c\text{-C}_4\text{H}_4\text{O}$ ($\kappa = 0.916$), and pyrrole $c\text{-C}_4\text{H}_4\text{NH}$ ($\kappa = 0.944$), has already been studied some time ago [30]. A more complete analysis is repeated here and the results are given in Table 5. In both cases, the A-reduction in representation III' gives the worst results and the best results are obtained for the S-reduction in representation III'. Both reductions in representation I' give almost identical results, the A-reduction being perhaps slightly better because the condition number is smaller. Another interesting case is nitric acid, HNO₃, whose rotational spectrum has been thoroughly studied [2]. We report in Table 5, the results of the analysis of the ground state of H¹⁵NO₃ [3] which is more asymmetric, $\kappa = 0.729$ (the advantage of the ¹⁵N species is that its spectrum is not complicated by the ¹⁴N quadrupole hyperfine structure). Here again, the A-reduction in representation III' gives the worst results but, contrary to the previous molecules, the S-reduction in representation III' is not com-

Table 5
Results of the centrifugal distortion analysis for a few oblate molecules.

Molecule		A-reduction		S-reduction	
		I'	III'	I'	III'
Furane	s_{111}	1.1×10^{-07}	-8.2×10^{-07}	-1.5×10^{-07}	-5.0×10^{-09}
	η^a	233	167	329	145
	s^b	30	83	31	22
	$s(\text{MAD})^c$	23	46	25	17
	P_{worst}^d	Φ_{JK} (37.6)	Φ_{KJ} (40)	Φ_{JK} (36.2)	Φ_{KJ} (20.0)
	p^e				
Pyrrole	s_{111}	1.1×10^{-07}	-6.2×10^{-07}	-1.3×10^{-07}	-4.8×10^{-09}
	η^a	187	294	327	67
	s^b	52	66	52	48
	$s(\text{MAD})^c$	36	47	36	38
	P_{worst}^d	Φ_{JK} (14.0)	Φ_{KJ} (13.3)	Φ_{JK} (13.2)	Φ_{KJ} (4.4)
	p^e				
H ¹⁵ NO ₃	s_{111}	4.6×10^{-07}	-1.6×10^{-06}	-5.0×10^{-07}	2.3×10^{-08}
	η^a	445	2197	573	350
	s^b	81	323	83	101
	$s(\text{MAD})^c$	53	137	55	54
	P_{worst}^d	ϕ_J (4.4)	ℓ_{JK} (1.9)	ℓ_3 (6.3)	L_K (18.9)
	p^e	21	22	22	22

^a Condition number, Eq. (A2).

^b Unweighted standard deviation of the fit in kHz.

^c Standard deviation of the fit, in kHz, calculated from the median absolute value, Eq. (A7).

^d Least well determined parameter, in parentheses, value of the parameter over its standard deviation.

^e Number of free parameters in the fit.

Table 6

Results of the centrifugal distortion analysis for a few prolate molecules (representation I').

Molecule		A-reduction	S-reduction
CH ₂ =CHCN	s_{111}	2.7×10^{-07}	2.1×10^{-08}
	η^a	2130	727
	s^b	12.7	13.7
	$s(\text{MAD})^c$	16.4	11.3
	P_{worst}^d	Φ_{JK} (8.6)	L_J (5.9)
	p^e	20	22
H ₂ CO	s_{111}	6.0×10^{-07}	-1.3×10^{-06}
	η^a	379	228
	s^b	265	221
	s_u^f	0.654	0.616
	P_{worst}^d	L_{KKJ} (16.9)	ℓ_4 (14.2)
	p^e	18	16

^a Condition number, Eq. (A2).

^b Unweighted standard deviation of the fit in kHz.

^c Standard deviation of the fit, in kHz, calculated from the median absolute value, Eq. (A7).

^d Least well determined parameter, in parentheses, value of the parameter over its standard deviation.

^e Number of free parameters in the fit.

^f Standard deviation of unit weight.

pletely satisfactory although it has the smallest s_{111} . The standard deviation of the fit is large and there are 15 lines with residuals larger than 300 kHz although the number of free parameters is large. The S-reduction in representation I' is slightly better but, surprisingly, the best results are obtained with the A-reduction in representation I': the standard deviation of the fit is the smallest one as well as the number of free parameters. Furthermore, as the sextic centrifugal distortion constant ϕ_J is not precisely determined, it is also possible to fix it without deteriorating the fit.

The situation for the prolate molecules is much simpler because s_{111} is at least one order of magnitude larger for the representation III' than for the representation I'. Thus, the choice is limited to the reduction. For instance, for a near-prolate top as vinyl cyanide, CH₂=CHCN, where $\kappa = -0.980$, $s_{111}(\text{A-I}') = 2.7 \times 10^{-7}$ is about 10 times larger than $s_{111}(\text{S-I}') = 2.1 \times 10^{-8}$ [28,31]. Without surprise, the S-reduction gives a better fit with a smaller condition number and a smaller standard deviation (with a smaller number of free parameters), see Table 6. The case of formaldehyde, H₂CO is more interesting. It is also a near-prolate molecule with $\kappa = -0.961$ but here $s_{111}(\text{A-I}') = 6.0 \times 10^{-7}$ is slightly smaller than $s_{111}(\text{S-I}') = -1.3 \times 10^{-6}$ [32]. However, the fit is still slightly better in the S-reduction with a smaller condition number and a smaller standard deviation (with moreover a smaller number of free parameters), see Table 6.

The first conclusion is that the A-reduction together with the representation III' is often a poor choice. Otherwise, the choice of the right Hamiltonian is not obvious and the Rays's asymmetry parameter is not a good indicator. It is much better to first calculate s_{111} from an *ab initio* harmonic force field (which is rather easy nowadays), to calculate the condition number for each fit, to check whether the parameters are precisely determined and to inspect the residuals to search for outliers or systematic deviations.

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Appendix A

The molecular parameters $\beta_j, j = 1, \dots, p$, are usually obtained by the least-squares method after linearization of the equations. The starting equation of the linear least-squares method is [33]

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (\text{A1})$$

where $\mathbf{y}$ is the vector of the input data, y_i , $i = 1, \dots, n$; $\mathbf{X}$ the Jacobian matrix whose elements are $X_{ij} = \partial v_i / \partial \beta_j$; and $\boldsymbol{\varepsilon}$ the vector of random errors.

Condition. number and ill-conditioning

The columns of the Jacobian matrix $\mathbf{X}$ are scaled to have unit length. The singular values μ_j (i.e. the square root of the eigenvalues of the matrix $\mathbf{X}^T \mathbf{X}$ where T means transpose) are calculated and the condition number is defined as the ratio of the largest singular value over the smallest one [28,34]

$$\eta = \frac{\mu_{\max}}{\mu_{\min}} \quad (\text{A2})$$

The usual notation for the condition number is κ but, to avoid any confusion with Ray's asymmetry parameter, we use η . A large condition number indicates a potential ill-conditioning. For Belsley [34], the cut-off value is 30 but, in general, a value of a few hundreds is still acceptable in a centrifugal distortion analysis.

Leverage. [27,28]

The observed vector $\mathbf{y}$ is transformed into its least-squares estimate $\hat{\mathbf{y}}$ by

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H}\mathbf{y} \quad (\text{A3})$$

The “hat” matrix $\mathbf{H}$ is a projection matrix, it is symmetric and idempotent and the following inequality holds for its diagonal terms h_{ii}

$$0 \leq h_{ii} \leq 1 \quad (\text{A4})$$

Eq. (A3) may be developed

$$\hat{y}_i = h_{i1}y_1 + \dots + h_{ii}y_i + \dots + h_{in}y_n \quad (\text{A5})$$

If $h_{ii} \approx 1$, then $\hat{y}_i \approx y_i$ and the corresponding residual will remain close to zero whatever the value of y_i . It happens in particular when one parameter is mostly determined by one single observation. It is obviously essential to avoid such influential data with large values of h_{ii} . Here again, the cut-of value is fuzzy. Some authors recommend $2p/n$ but this is often too strict and data for which $h_{ii} < 0.6$ are rather safe.

Iteratively. reweighted least-squares method [29]

The method involves the following steps:

Step 0. Initial residuals $r_i = v_i^{\text{exp}} - v_i^{\text{calc}}$ of the fit are calculated using the ordinary least-squares method.

Step 1. The median of absolute deviations (MAD) of the residuals is calculated

$$\text{MAD} = \text{median}|r_i - \text{median}(r_i)| \quad (\text{A6})$$

and the standard deviation is estimated from MAD as

$$s = 1.4826 \left(1 + \frac{5}{n-p} \right) \text{MAD} \quad (\text{A7})$$

where n is the number of data and p the number of fitted parameters. $5(n-p)$ is a finite sample correction factor which is important for small samples. This estimation should be close to the value of the usual standard deviation but, contrary to the usual definition of the standard deviation, it is robust, i.e. not sensitive to outliers. When the errors are normally distributed, Eq. (A7) is an estimate of the population standard deviation. A large difference between the usual standard deviation and the result of Eq. (A7) indicates a problem in the fit. The residuals are scaled by dividing them by s from Eq. (A7)

$$u_i = |r_i|/s \quad (\text{A8})$$

and the weights are calculated using the biweight function

$$\begin{cases} w_i = w_i^H \left[1 - \left(\frac{u_i}{4.685} \right)^2 \right]^2 & \text{if } u_i \leq 4.685 \\ w_i = 0 & \text{if } u_i > 4.685 \end{cases} \quad (\text{A9})$$

where

$$\begin{cases} w_i^H = 1 & \text{if } h_{ii} \leq 0.3 \\ w_i^H = \left(\frac{0.3}{h_{ii}} \right)^2 & \text{if } h_{ii} > 0.3 \end{cases} \quad (\text{A10})$$

The weights w_i^H are introduced to reduce the influence of leverage points.

Step 2. The weighted least-squares method is used with these weights, Eq. (A9), to obtain a new set of parameters.

Step 3. The processus, steps 1 and 2, is repeated until there is negligible change from one iteration to the next. Note that, during the iterations, some lines may be excluded from the fit.

Appendix B. Supplementary data

Supplementary data for this article are available on Science Direct (www.sciencedirect.com) and as part of the Ohio State University Molecular Spectroscopy Archives (http://library.osu.edu/sites/msa/jmsa_hp.htm).

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